

Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number: 13

Start Date: 02-02-2026

Name: Jaya Varsini M

Internship Duration: 8 Weeks

1. Project Title

TalkMetrix – GenAI-Powered Customer Support Quality Auditor

2. Project Objective

To develop an AI-powered system that evaluates customer support interactions, detects compliance violations, and improves service quality using LLMs and RAG.

3. Project Description

- Designed a full-stack web application for analyzing customer support conversations
- Implemented file upload functionality for chat logs and audio files
- Integrated speech-to-text transcription using Whisper (with fallback handling)
- Developed an LLM-based scoring engine using Groq (Llama3)
- Implemented evaluation metrics:
 - Empathy
 - Compliance
 - Resolution
 - Professionalism
- Built a RAG (Retrieval-Augmented Generation) pipeline using FAISS for policy-based evaluation
- Created dashboards for agent performance and analytics
- Developed alert systems for compliance violations
- Enabled report generation (PDF and Excel)

Technologies Used:

Frontend:

- React (TypeScript, Vite)
- Tailwind CSS, shadcn/ui
- Recharts (analytics visualization)

Backend:

- FastAPI (Python)
- SQLite database

AI/ML:

- Groq LLM (Llama3-70B)
- Whisper (Speech-to-Text)
- FAISS (RAG implementation)

Deployment:

- Netlify (Frontend)
- Render (Backend)

Real-World Applications:

- Automates customer support quality audits
- Reduces manual QA workload
- Ensures compliance with company policies
- Improves customer experience
- Scalable for enterprise-level usage

4. Timeline Overview

Week	Planned Activities	Completed Work
Week 1	Project planning and research	Understood GenAI, QA systems, finalized project scope
Week 2	Frontend setup	Built React UI and navigation
Week 3	Backend setup	Developed FastAPI APIs
Week 4	File upload + transcription	Implemented upload system and Whisper integration
Week 5	LLM scoring engine	Implemented scoring metrics and AI suggestions
Week 6	RAG + dashboards	Integrated FAISS and built analytics dashboard
Week 7	Alerts + reports	Developed alert system and report generation
Week 8	Testing & deployment	Deployed project on Netlify and Render

5a. Key Milestones

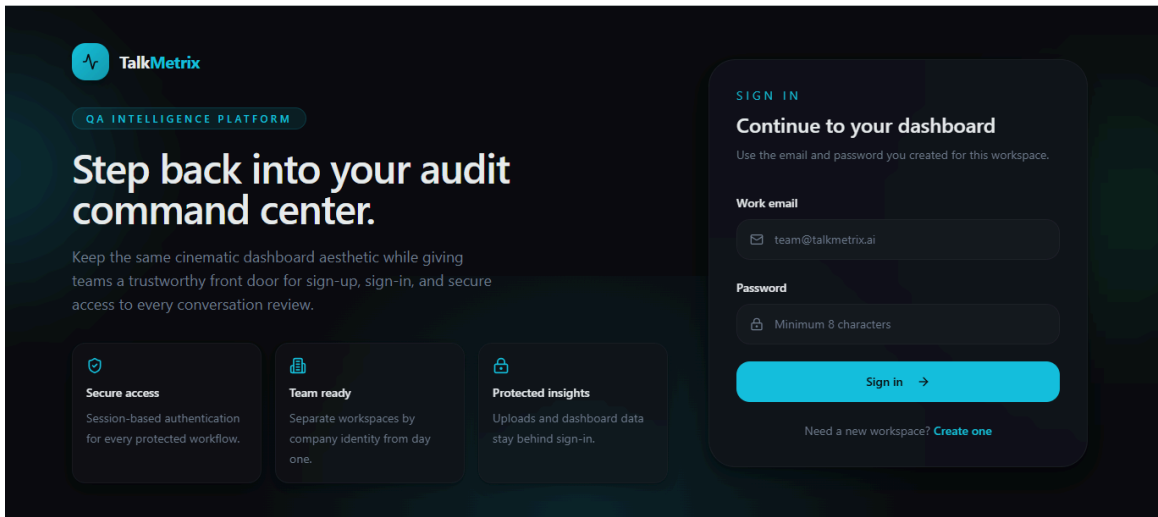
Milestone	Description	Completion
Project Kickoff	Planning and requirement analysis	Week 1
Prototype	Initial ML model and UI ready	Week 3
Mid-Term Review	LLM scoring and integration	Week 5
Final Submission	Fully functional system deployed	Week 6
Presentation	Final demonstration and documentation	Week 7

5b. Project Execution Details

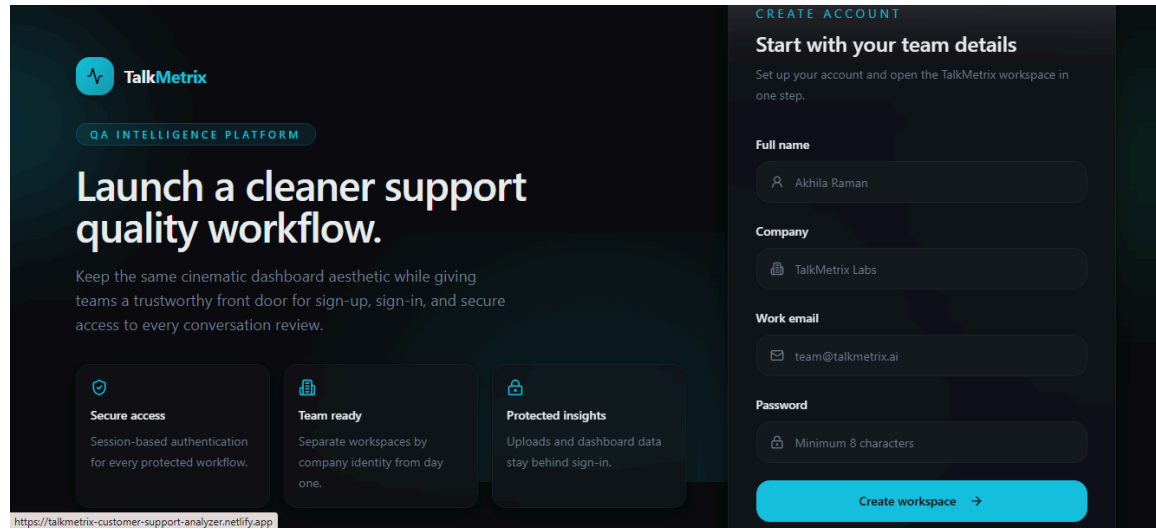
- The project started with requirement analysis and research on customer support quality auditing systems. A full-stack architecture was designed using React for frontend and FastAPI for backend.
- Initially, file upload functionality was implemented to process chat and audio data. Whisper-based transcription was integrated to convert audio into text. The core of the system involved building an LLM-based scoring engine using Groq, which evaluates conversations across multiple quality parameters.
- To enhance contextual understanding, a RAG pipeline using FAISS was implemented to retrieve policy-related information. The frontend dashboard was developed to visualize agent performance using charts and analytics.
- Further enhancements included implementing compliance alerts and generating detailed reports in PDF and Excel formats. Finally, the system was tested and deployed successfully for real-time usage.

6. Snapshots

A) Login Page



B) Create Account



The 'CREATE ACCOUNT' page features a dark theme. On the left, the TalkMetrix logo is at the top, followed by the tagline 'QA INTELLIGENCE PLATFORM'. Below this is a large heading 'Launch a cleaner support quality workflow.' and a paragraph explaining the platform's benefits. Three feature boxes are listed: 'Secure access' (session-based authentication), 'Team ready' (separate workspaces), and 'Protected insights' (data security). On the right, the 'Start with your team details' section includes input fields for 'Full name' (Akhila Raman), 'Company' (TalkMetrix Labs), 'Work email' (team@talkmetrix.ai), and 'Password' (with a strength indicator). A 'Create workspace' button is at the bottom right.

TalkMetrix
QA INTELLIGENCE PLATFORM

Launch a cleaner support quality workflow.

Keep the same cinematic dashboard aesthetic while giving teams a trustworthy front door for sign-up, sign-in, and secure access to every conversation review.

- Secure access**
Session-based authentication for every protected workflow.
- Team ready**
Separate workspaces by company identity from day one.
- Protected insights**
Uploads and dashboard data stay behind sign-in.

CREATE ACCOUNT
Start with your team details
Set up your account and open the TalkMetrix workspace in one step.

Full name
Akhila Raman

Company
TalkMetrix Labs

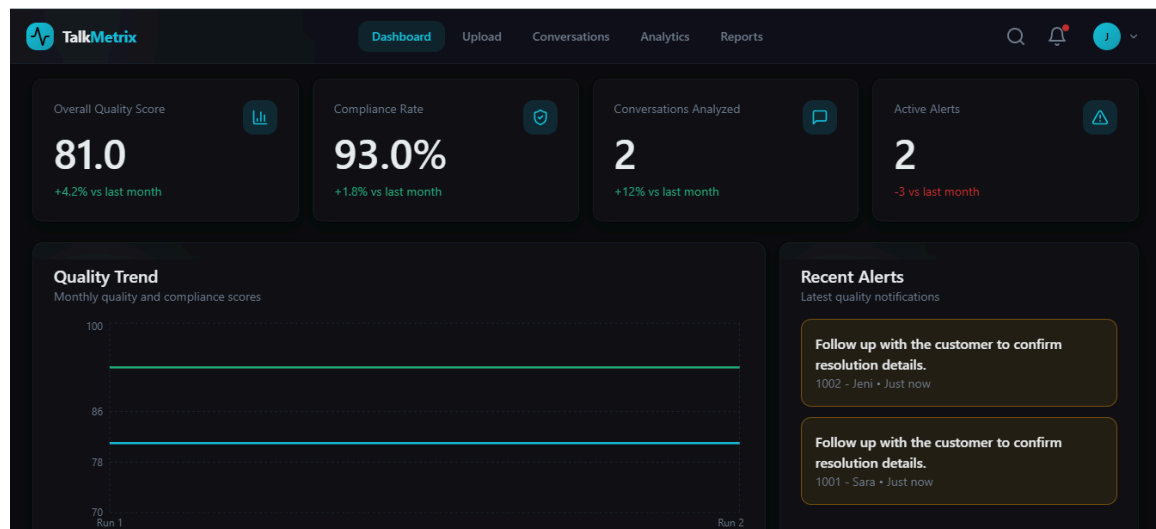
Work email
team@talkmetrix.ai

Password
Minimum 8 characters

Create workspace →

<https://talkmetrix-customer-support-analyzer.netlify.app>

C) Dashboard



D)Upload page

TalkMetrix

DashboardUploadConversationsAnalyticsReports

Agent ID

AG-204

Agent Name

Akhila Raman

Drop files here or click to browse

Supports .txt, .pdf, and audio files like .mp3, .wav, .m4a, .aac, .ogg, and more

Select File

No file selected yet

Upload and Analyze

E)Conversations(Transcription)

TalkMetrix

DashboardUploadConversationsAnalyticsReports

Search conversations...

CONV-BFBCCE81

1002 - Jeni

Automated local scoring completed.

2026-03-27 • 00:00

Delete

CONV-81782181

1001 - Sara

Automated local scoring completed.

2026-03-27 • 00:00

Delete

CONV-BFBCCE81

1002 - Jeni -> Customer

2026-03-27 - 00:00 - Live Chat

Transcript

Agent - 0:00

Hi, this is Maria Rodriguez. I'm having trouble with my LAP2110 laptop, order number 987654. The Wi-Fi keeps disconnecting, and I can't get a stable internet connection.

Customer - 0:08

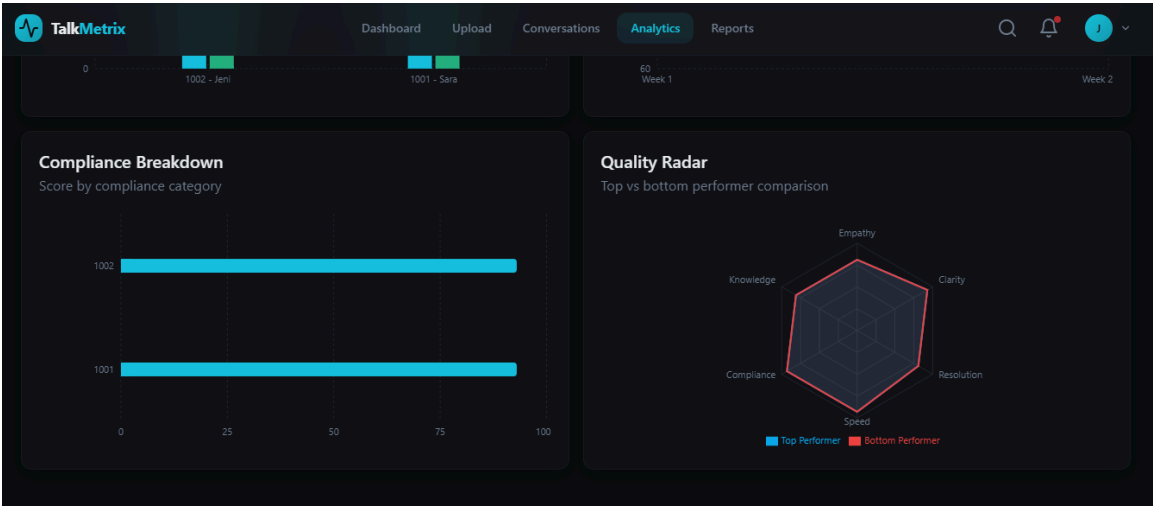
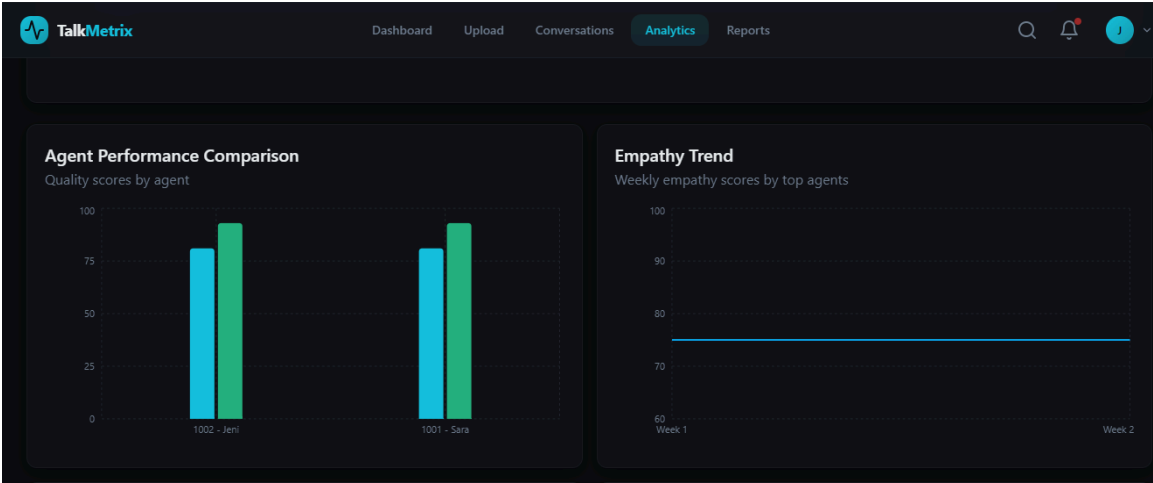
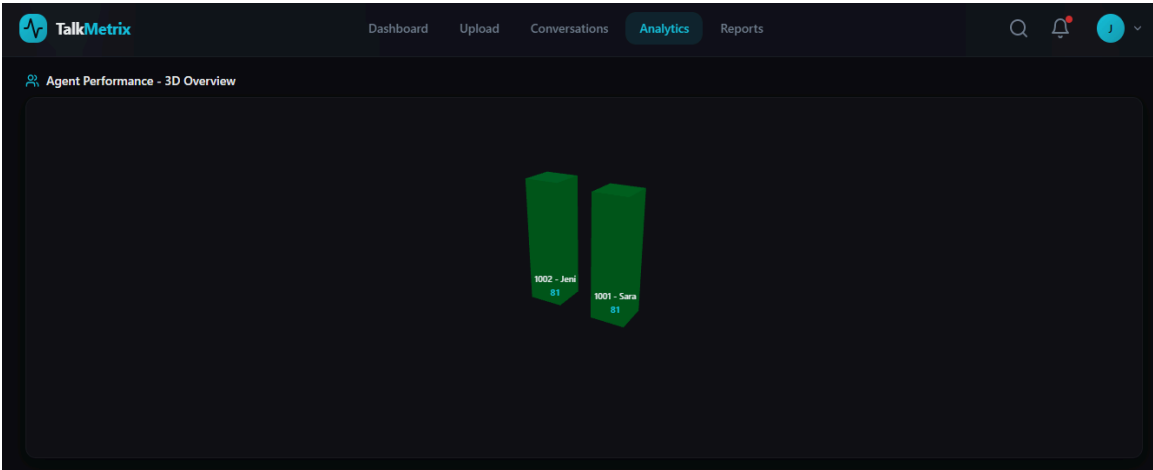
I've tried restarting it multiple times, and I've checked my router, but everything else is working fine. This is incredibly frustrating. I need this for work.

Agent - 0:16

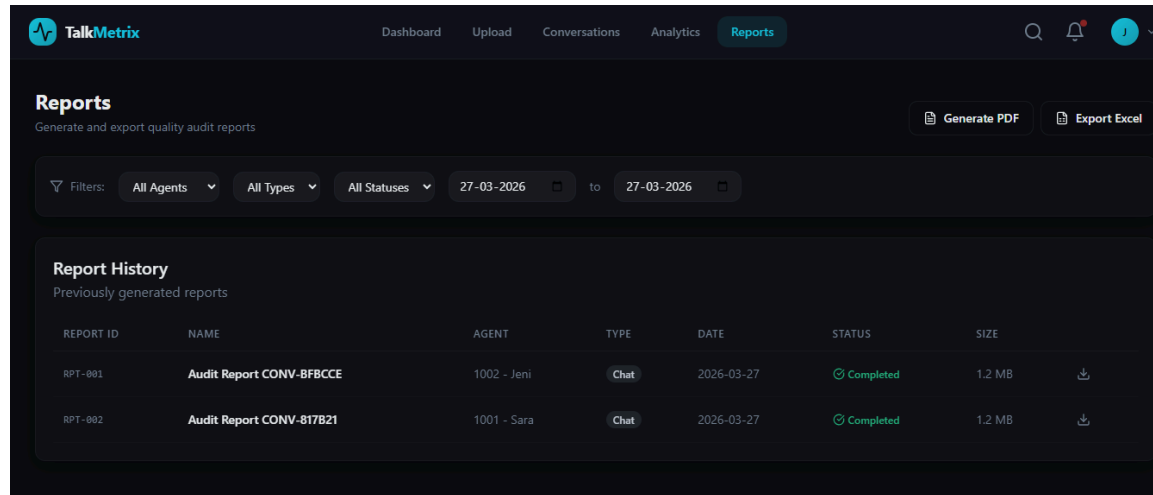
Can you please help me troubleshoot this?

AI Suggestions

F)Analytics



G)Reports



7. Challenges Faced

- **Whisper Transcription Issues:** Audio processing inconsistencies resolved using fallback mechanisms
- **Prompt Engineering:** Difficulty in getting structured LLM output solved through prompt refinement
- **RAG Integration Complexity:** Managed using FAISS with optimized retrieval logic
- **Frontend-Backend Integration:** Fixed API communication and data formatting issues
- **Deployment Issues:** Resolved environment and dependency conflicts .

8. Learnings & Skills Acquired

Technical Skills:

- Full-stack development using React and FastAPI
- LLM integration (Groq / OpenAI concepts)
- RAG pipeline using FAISS
- Speech-to-text processing (Whisper)
- API development and deployment
- Dashboard visualization using Recharts

Soft Skills:

- Problem-solving and debugging
- Time management
- Agile development practices

9. Testimonials

This project provided valuable hands-on experience in building a real-world AI-powered application. It enhanced my understanding of GenAI, full-stack development, and system integration. Successfully developing and deploying a complete customer support auditing system was a significant achievement during this internship.

10. Conclusion

The internship project successfully demonstrated the application of Generative AI in automating customer support quality auditing. It provided practical exposure to integrating AI models, building scalable web applications, and solving real-world problems. This experience aligns strongly with my career goals in AI and software development.

11. Acknowledgements

I would like to express my sincere gratitude to my mentor and the Infosys Springboard team for their continuous guidance and support throughout this internship. I also thank my peers and contributors who helped in completing this project successfully.